# FRANCES (SU) LIU

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## RESEARCH INTERESTS

My research asks when and how different information sets sharpen volatility forecasts, combining high-frequency market microstructure data and media sentiment within a common HAR-based framework. Building on this and on six years of prior industry experience in equity strategy and portfolio management, I am also interested in extending these methods to high-dimensional portfolio allocation, combining dependence modeling with machine learning.

## EDUCATION

**Doctor of Philosophy (PhD) in Finance** *2024.02–2027.08 (expected)*
*University of Technology Sydney (UTS), Sydney, Australia*

*Supervisors*: Assoc. Prof. Vitali Alexeev (UTS, Finance) & Assoc. Prof. Katja Ignatieva (UNSW, Actuarial)

**Master of Science in Financial Engineering** *2015.08–2017.06*
*Stevens Institute of Technology, New Jersey, United States* *GPA: 3.9/4.0*

*Thesis*: Systemic Risk Measurement: CoVaR and Copula Models (ARMA/TGARCH conditional moments, copula-based dependence between financial sectors, time-series CoVaR). Supervised by Assoc. Prof. Zhenyu Cui (School of Systems & Enterprises, Hanlon Lab)

**Bachelor of Economics in Financial Engineering** *2011.09–2015.06*
*Harbin University of Commerce, Harbin, China*

## WORKING PAPERS

**Reading Between the Trades: Neural Encoding Extended-Hours Tick Sequences for Volatility Forecasting**
with Vitali Alexeev (UTS), Adam Clements (QUT), and Katja Ignatieva (UNSW) *working paper*

- Replaces the coarse overnight aggregates used in the volatility literature with the complete tick-level record of each extended-hours session, compressing variable-length, irregularly spaced transaction sequences into fixed-dimensional representations through neural encoders and integrating them with the HAR model
- An incremental architecture hierarchy isolates the contribution of each modeling ingredient; applied to all DJIA constituents over 2002–2025, the simplest architecture reduces out-of-sample forecast loss by approximately 11% at the one-day horizon, roughly twice the gain of the best scalar overnight proxy
- Pre-open and post-close sessions deliver comparable aggregate gains through opposite channels: 64% of the pre-open improvement is recoverable from a single within-session aggregate, while 81% of the post-close improvement is attributable to tick-sequence structure. Gains are stable across the Global Financial Crisis and COVID-19 episodes
- *JEL*: C45, C53, G12, G14

**When Does Sentiment Predict Volatility?**
with Vitali Alexeev (UTS) and Katja Ignatieva (UNSW) *under review, Journal of Applied Econometrics*

- Augmenting HAR models with Thomson Reuters MarketPsych sentiment yields only modest average improvements, but this average masks strong state dependence
- Sentiment becomes materially more informative in the upper tail of the volatility distribution and, more importantly, when volatility persistence weakens; persistence is measured within a fractional integration framework
- Out-of-sample gains concentrate in regimes jointly characterised by high volatility and low persistence, indicating that sentiment supplies forward-looking narrative information precisely when price-based models are least reliable
- *JEL*: C22, C53, G12, G14

## CONFERENCE PRESENTATIONS

*When Does Sentiment Predict Volatility?*

- Econometric Society Australasia Meeting (ESAM), Adelaide, Australia, November 2026 *(accepted)*
- FIRN Annual Conference, K'gari, Australia, November 2026 *(accepted)*
- International Association for Applied Econometrics (IAAE) Annual Conference, Carcavelos, Portugal, June 2026

*Reading Between the Trades*

- Econometric Society Australasia Meeting (ESAM), Adelaide, Australia, November 2026 *(accepted)*

## AWARDS AND SCHOLARSHIPS

UTS Business Doctoral Scholarship (full stipend) 2024–2027
UTS International Research Scholarship (full tuition) 2024–2027
Academic Excellence Award, Stevens Institute of Technology 2017

## TEACHING EXPERIENCE

**University of Technology Sydney** *Sydney, Australia*
*Tutor, Finance Discipline Group* 2024–present

- 25503 *Investment Analysis* (undergraduate), Spring 2026
- 25556 *The Financial System* (undergraduate), Autumn 2026 and Spring 2026

## INDUSTRY EXPERIENCE

Six years of applied quantitative research, carrying models from hypothesis through backtesting to deployment.

**Webull Financial LLC** *Changsha, China*
*Senior Quantitative Analyst* 2020.04–2023.11

- Built the portfolio construction behind a mass-market advised-portfolio product: mean-variance allocation with CAPM reverse-optimized returns, machine-learning denoising, and shrinkage-based covariance estimation
- Delivered a factor-driven stock selection product for US equities, from factor research on S&P, FactSet, and Refinitiv data through to production
- Developed multivariate time-series and factor models for private-fund strategies across Chinese and US equity index universes, validated and monitored after deployment
- Designed and co-developed the quantitative research database and strategy backtesting environment used across the quantitative team

**Pricing and Analytics (Shenzhen) Limited** *Shenzhen, China*
*Quantitative Analyst: Derivatives Pricing and Valuation* 2017.12–2019.10

- Designed pricing and valuation models for mortgage-backed products including REITs and CLOs; both models were registered as software copyrights
- Applied Black-Scholes, risk-neutral valuation, and martingale pricing methods, and implemented extreme value theory and Value at Risk for portfolio tail risk

## TECHNICAL SKILLS

**Programming:** Python (PyTorch, pandas, NumPy), R, C++, SQL
**Econometrics:** time-series and realized volatility modeling on tick-level data, forecast evaluation and comparison, quantile methods, long-memory and fractional integration
**Machine learning:** deep sequence models (attention, convolutional, recurrent, set-based); out-of-sample validation and model selection
**Quantitative finance:** portfolio construction and optimization, dependence modeling with copulas, tail risk (extreme value theory, VaR, CVaR), derivatives pricing and valuation
**Computing:** GPU cluster workloads; reproducible large-scale experiment pipelines
**Languages:** English (proficient), Cantonese (proficient), Mandarin (native)